Google ADK Multi-Agent Development to Deployment Guide

1. Goal

Build an Investment Advisor AI where every request flows: User → Root Agent → Intent Agent → Investment Agent (Sequential) → Retrieval → Recommendation → Compliance, or to General Helper Agent.

2. Recommended Architecture

User
↓
Root Agent
↓
Intent Agent
■■ Investment Agent (Sequential)
■ ■■ Retrieval Agent
■ ■■ Recommendation Agent
■ ■■ Compliance Agent
■■ General Helper Agent

3. Parent Rule

Each ADK agent can have only one parent. Never reuse the same agent object in two different sub_agents lists. If needed, redesign the hierarchy instead of sharing agent instances.

4. Common Error

ValidationError: Agent already has a parent agent. Cause: the same agent (for example retrieval_agent) was added under both intent_agent and investment_agent. Fix: keep retrieval_agent only inside investment_agent.

5. Function Calling vs output_schema

Do not combine output_schema with sub_agents/tools on the same LlmAgent. Remove output_schema from routing agents that delegate to sub-agents.

6. Suggested Agents

Root Agent: receives user request and delegates to Intent Agent.
Intent Agent: classifies request and delegates to Investment Agent or General Helper Agent.
Investment Agent (Sequential): Retrieval → Recommendation → Compliance.
General Helper Agent: greetings and generic questions.

7. Folder Structure

investmentAI/
■■■ agent.py
■■■ agents/
■ ■■■ intent_agent.py
■ ■■■ general_helper_agent.py
■ ■■■ investment/
■ ■■■ investment_agent.py
■ ■■■ retrieval_agent.py

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■       ■■■ recommendation_agent.py
■       ■■■ compliance_agent.py
■■■ instructions/
■■■ schemas/
■■■ tools/
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8. Development Lifecycle

Requirements → Design → Build Agents → Create Prompts/Instructions → Connect Tools → Test Locally → Evaluate → Deploy → Log → Monitor → Improve & Redeploy

9. Evaluation

Measure intent accuracy, retrieval quality, faithfulness, answer relevancy, context precision, context recall, latency, token usage, and failure rate. Use RAG evaluation before production.

10. Logging

Capture user queries, agent routing decisions, tool invocations, latency, tokens, errors, and final responses. Store logs in systems such as BigQuery or Cloud Logging.

11. Monitoring

Track latency, cost, routing failures, hallucinations, retrieval success, API failures, and user feedback. Configure alerts for abnormal behavior.

12. Deployment

Run locally with ADK, containerize if needed, deploy on Google Cloud (Cloud Run/GKE/Vertex AI depending on architecture), secure secrets, enable logging and monitoring.

13. Production Best Practices

Use one parent per agent, modular prompts, version control, automated evaluation, retries, observability, and continuous improvement.